PH-201: Electricity & Magnetism - I (Theory)

Lecture Note 3 ver: 1.0.0

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1. Introduction

We have already seen in provious lectures that we can use Couloumb's law to calculate the electric field, potential, and electrostatic force acting on a given charge due to a simple point charge or a complex collection of charges or a large charged body with a given charge density function ρ. In this class we are going to explore **Gauss's Law**, it provides an alternative approach for calculating the previous quantities.

Gauss's Law states that the net electric flux through any closed surface is directly proportional to the net electric charge enclosed within the surface, irrespective of how the charge is distributed or the shape of the surface. Considering an electric field at a position $\vec{\boldsymbol{r}}$, $\overrightarrow{\boldsymbol{E}_{\boldsymbol{\rho}}}(\vec{\boldsymbol{r}})$ generated by a body with a charge density ρ, Gauss's law states that the total electric flux through a surface S enclosing the charged body, $\oiint_S \overrightarrow{\boldsymbol{E}_{\boldsymbol{\rho}}}(\vec{\boldsymbol{r}}) \cdot \mathrm{d}\vec{\boldsymbol{a}}(\vec{\boldsymbol{r}})$ follows

$$\oiint_S \overrightarrow{\boldsymbol{E}_{\boldsymbol{\rho}}}(\vec{\boldsymbol{r}}) \cdot \mathrm{d}\vec{\boldsymbol{a}}(\vec{\boldsymbol{r}}) = \iiint_V \frac{\rho(\vec{\boldsymbol{r}}')}{\varepsilon_0} \mathrm{d}^3\vec{\boldsymbol{r}}' \tag{1}$$

There are two forms of the equation describing Gauss's law, an integral form and a differential form. Equation 1 represents the integral form of Gauss's law. We will first derive the integral form for various configurations of charges in vacuum and then we will derive the differential form in the next section.

2. Integral Form of Gauss's Law

Let us now consider a closed surface S, enclosing a volume V, and a point charge, Q_A, at an arbitrary point A inside this enclosed volume. The electric field going through a small portion of this surface S at position $\vec{\boldsymbol{r}}_{\boldsymbol{B}}$ can be represented by $\overrightarrow{\boldsymbol{E}_{\boldsymbol{Q_A}}}(\vec{\boldsymbol{r}}_{\boldsymbol{B}})$. The flux passing through this small portion $\mathrm{d}\vec{\boldsymbol{a}}(\vec{\boldsymbol{r}}_{\boldsymbol{B}})$ is $\overrightarrow{\boldsymbol{E}_{\boldsymbol{Q_A}}}(\vec{\boldsymbol{r}}_{\boldsymbol{B}}) \cdot \mathrm{d}\vec{\boldsymbol{a}}(\vec{\boldsymbol{r}}_{\boldsymbol{B}})$. Figure 1 illustrates the surface S, the point charge Q_A, and the infinitesimally small portion of S, $\mathrm{d}\vec{\boldsymbol{a}}$ clearly for your consideration. To derive the integral form of Gauss's law we need to derive the expression for the total flux passing through this closed surface S, i.e. derive an expression for $\oiint_S \overrightarrow{\boldsymbol{E}_{\boldsymbol{Q_A}}}(\vec{\boldsymbol{r}}_{\boldsymbol{B}}) \cdot \mathrm{d}\vec{\boldsymbol{a}}(\vec{\boldsymbol{r}}_{\boldsymbol{B}})$.

Now, using Columb's law we can write the flux passing through $\mathrm{d}\vec{\boldsymbol{a}}(\vec{\boldsymbol{r}}_{\boldsymbol{B}})$ as

$$\begin{aligned} \overrightarrow{\boldsymbol{E}_{\boldsymbol{Q_A}}}(\vec{\boldsymbol{r}}_{\boldsymbol{B}}) \cdot \mathrm{d}\vec{\boldsymbol{a}}(\vec{\boldsymbol{r}}_{\boldsymbol{B}}) &= \frac{1}{4\pi\varepsilon_0} Q_A \frac{\vec{\boldsymbol{r}}_{\boldsymbol{B}} - \vec{\boldsymbol{r}}_{\boldsymbol{A}}}{\left|\vec{\boldsymbol{r}}_{\boldsymbol{B}} - \vec{\boldsymbol{r}}_{\boldsymbol{A}}\right|^3} \cdot \mathrm{d}\vec{\boldsymbol{a}}(\vec{\boldsymbol{r}}_{\boldsymbol{B}}) \\ &= \frac{1}{4\pi\varepsilon_0} Q_A \frac{\cos(\theta)\, \mathrm{d}a(\vec{\boldsymbol{r}}_{\boldsymbol{B}})}{\left|\vec{\boldsymbol{r}}_{\boldsymbol{B}} - \vec{\boldsymbol{r}}_{\boldsymbol{A}}\right|^2} \end{aligned} \tag{2}$$

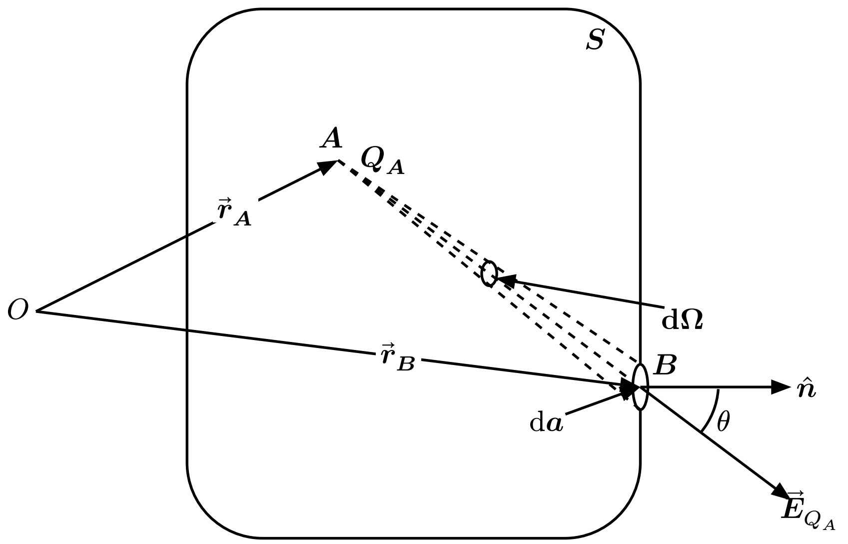

Figure 1: This figure illustrates a point charge, Q_A, at point A whose position is $\vec{r}_A$ w.r.t the origin O. The charge is inside the volume enclosed by the surface S. The electric field due to the point charge Q_A at point B is represented by $\vec{E}_{Q_A}$. The infinitesimally small portion of the surface at position $\vec{r}_B$ is illustrated as da with the surface normal $\hat{n}$. The solid angle covered by this infinitesimally small area at point A is represented by d$\boldsymbol{\Omega}$.

Let us use d$\boldsymbol{\Omega}$ to represent the solid angle at point A that the infinitesimally small area $\mathrm{d}\vec{a}(\vec{r}_B)$ covers. Now, d$\boldsymbol{\Omega}$ can also be written in terms of the component of the area in line with $\vec{r} = \vec{r}_B - \vec{r}_A$ and $|\vec{r}|$ as

$$\begin{aligned}\mathrm{d}\boldsymbol{\Omega} &= \frac{\text{infinitesimally small area component}}{\text{square of radius}} \\ &= \frac{(\cos(\theta) \times \mathrm{d}a)}{|\vec{r}|^2}\end{aligned}$$

Therefore, the flux flowing through $\mathrm{d}\vec{a}(\vec{r}_B)$ can be written in terms of d$\boldsymbol{\Omega}$ as

$$\begin{aligned}\vec{E}_{Q_A}(\vec{r}_B) \cdot \mathrm{d}\vec{a}(\vec{r}_B) &= \frac{1}{4\pi\varepsilon_0} Q_A \frac{\cos(\theta)\,\mathrm{d}a(\vec{r}_B)}{|\vec{r}_B - \vec{r}_A|^2} \\ &= \frac{1}{4\pi\varepsilon_0} Q_A\,\mathrm{d}\boldsymbol{\Omega}\end{aligned}$$

Now, to calculate the total flux through the closed surface S we need to integrate $\vec{E}_{Q_A}(\vec{r}_B) \cdot \mathrm{d}\vec{a}(\vec{r}_B)$ over the complete closed surface S. This will lead to the following

$$\begin{aligned}\oiint_S \vec{E}_{Q_A}(\vec{r}_B) \cdot \mathrm{d}\vec{a}(\vec{r}_B) &= \oiint_S \frac{1}{4\pi\varepsilon_0} Q_A \frac{\vec{r}_B - \vec{r}_A}{|\vec{r}_B - \vec{r}_A|^3} \cdot \mathrm{d}\vec{a}(\vec{r}_B) \\ &= \oiint_S \frac{1}{4\pi\varepsilon_0} Q_A \frac{\cos(\theta)\,\mathrm{d}a(\vec{r}_B)}{|\vec{r}_B - \vec{r}_A|^2} \\ &= \oiint_S \frac{1}{4\pi\varepsilon_0} Q_A \frac{\cos(\theta)\,\mathrm{d}a(\vec{r}_B)}{|\vec{r}_B - \vec{r}_A|^2} \\ &= \int_0^{4\pi} \frac{1}{4\pi\varepsilon_0} Q_A\,\mathrm{d}\boldsymbol{\Omega} = \frac{Q_A}{\varepsilon_0}\end{aligned}$$

We have thus derived the integral form of Gauss's law for a point charge inside an enclosed surface. We can now use similar methods to derive the integral form of Gauss's law for a collection of point charges $Q_i, i = 0, 1, ..., N$ placed at positions $\vec{\boldsymbol{r}}_{\boldsymbol{i}}, i = 0, 1, ..., N$ respectively. The derivation is as follows

$$
\begin{aligned}
\oiint_S \overrightarrow{\boldsymbol{E}}_{\{Q_i\}_1^N}(\vec{\boldsymbol{r}}_{\boldsymbol{B}}) \cdot \mathrm{d}\vec{\boldsymbol{a}}(\vec{\boldsymbol{r}}_{\boldsymbol{B}}) &= \oiint_S \left\{ \sum_{i=1}^{N} \overrightarrow{\boldsymbol{E}}_{Q_i}(\vec{\boldsymbol{r}}_{\boldsymbol{B}}) \right\} \cdot \mathrm{d}\vec{\boldsymbol{a}}(\vec{\boldsymbol{r}}_{\boldsymbol{B}}) \\
&= \oiint_S \frac{1}{4\pi\varepsilon_0} \left\{ \sum_{i=1}^{N} Q_i \frac{\vec{\boldsymbol{r}}_{\boldsymbol{B}} - \vec{\boldsymbol{r}}_{\boldsymbol{i}}}{\left|\vec{\boldsymbol{r}}_{\boldsymbol{B}} - \vec{\boldsymbol{r}}_{\boldsymbol{i}}\right|^3} \right\} \cdot \mathrm{d}\vec{\boldsymbol{a}}(\vec{\boldsymbol{r}}_{\boldsymbol{B}}) \\
&= \oiint_S \frac{1}{4\pi\varepsilon_0} \left\{ \sum_{i=1}^{N} \left(Q_i \frac{\cos(\theta)\, \mathrm{d}a(\vec{\boldsymbol{r}}_{\boldsymbol{B}})}{\left|\vec{\boldsymbol{r}}_{\boldsymbol{B}} - \vec{\boldsymbol{r}}_{\boldsymbol{i}}\right|^2} \right) \right\} \\
&= \sum_{i=1}^{N} \left\{ \oiint_S \frac{1}{4\pi\varepsilon_0} Q_i \frac{\cos(\theta)\, \mathrm{d}a(\vec{\boldsymbol{r}}_{\boldsymbol{B}})}{\left|\vec{\boldsymbol{r}}_{\boldsymbol{B}} - \vec{\boldsymbol{r}}_{\boldsymbol{i}}\right|^2} \right\} \\
&= \sum_{i=1}^{N} \left\{ \int_0^{4\pi} \frac{1}{4\pi\varepsilon_0} Q_i \, \mathrm{d}\boldsymbol{\Omega} \right\} \\
&= \sum_{i=1}^{N} \frac{Q_i}{\varepsilon_0}
\end{aligned}
$$

We can now similarly derive the integral form of Gauss's law for any arbitrary charged body inside the volume enclosed by the surface S. We consider a charge density function ρ which provides the charge density at any position inside the enclosed volume. The derivation is simple and shows that the configuration of the charge inside the enclosed surface is not important when calculating the total flux through the surface. The derivation is as follows

$$
\begin{aligned}
\oiint_S \overrightarrow{\boldsymbol{E}}_{\rho}(\vec{\boldsymbol{r}}_{\boldsymbol{B}}) \cdot \mathrm{d}\vec{\boldsymbol{a}}(\vec{\boldsymbol{r}}_{\boldsymbol{B}}) &= \oiint_S \left\{ \int \mathrm{d}\overrightarrow{\boldsymbol{E}}_{\rho}(\vec{\boldsymbol{r}}_{\boldsymbol{B}}) \right\} \cdot \mathrm{d}\vec{\boldsymbol{a}}(\vec{\boldsymbol{r}}_{\boldsymbol{B}}) \\
&= \oiint_S \frac{1}{4\pi\varepsilon_0} \left\{ \iiint_V \rho(\vec{\boldsymbol{r}}') \frac{\vec{\boldsymbol{r}}_{\boldsymbol{B}} - \vec{\boldsymbol{r}}'}{\left|\vec{\boldsymbol{r}}_{\boldsymbol{B}} - \vec{\boldsymbol{r}}'\right|^3} \mathrm{d}^3\vec{\boldsymbol{r}}' \right\} \cdot \mathrm{d}\vec{\boldsymbol{a}}(\vec{\boldsymbol{r}}_{\boldsymbol{B}}) \\
&= \oiint_S \frac{1}{4\pi\varepsilon_0} \left\{ \iiint_V \left(\rho(\vec{\boldsymbol{r}}') \frac{\cos(\theta)}{\left|\vec{\boldsymbol{r}}_{\boldsymbol{B}} - \vec{\boldsymbol{r}}'\right|^2} \right) \mathrm{d}^3\vec{\boldsymbol{r}}' \right\} \mathrm{d}a(\vec{\boldsymbol{r}}_{\boldsymbol{B}}) \\
&= \iiint_V \left\{ \oiint_S \frac{1}{4\pi\varepsilon_0} \rho(\vec{\boldsymbol{r}}') \frac{\cos(\theta)}{\left|\vec{\boldsymbol{r}}_{\boldsymbol{B}} - \vec{\boldsymbol{r}}'\right|^2} \, \mathrm{d}a(\vec{\boldsymbol{r}}_{\boldsymbol{B}}) \right\} \mathrm{d}^3\vec{\boldsymbol{r}}' \\
&= \iiint_V \left\{ \int_0^{4\pi} \frac{\rho(\vec{\boldsymbol{r}}')}{4\pi\varepsilon_0} \, \mathrm{d}\boldsymbol{\Omega} \right\} \mathrm{d}^3\vec{\boldsymbol{r}}' \\
&= \iiint_V \frac{\rho(\vec{\boldsymbol{r}}')}{\varepsilon_0} \mathrm{d}^3\vec{\boldsymbol{r}}'
\end{aligned}
$$

3. Differential Form of Gauss's Law

We have seen in the earlier section, the integral form of Gauss's Law. In this section we are going to derive the differential form of Gauss's Law. We begin with **Gauss's Divergence Theorem** which can be written as

$$\oint\!\!\!\oint_S \vec{\boldsymbol{A}}(\vec{\boldsymbol{r}}) \cdot \mathrm{d}\vec{\boldsymbol{a}}(\vec{\boldsymbol{r}}) = \iiint_V \overrightarrow{\boldsymbol{\nabla}}_{\vec{\boldsymbol{r}}} \cdot \vec{\boldsymbol{A}}(\vec{\boldsymbol{r}}) \mathrm{d}^3\vec{\boldsymbol{r}}$$

we can write $\mathrm{d}\vec{\boldsymbol{a}}(\vec{\boldsymbol{r}})$ as $\hat{n}\,\mathrm{d}a(\vec{\boldsymbol{r}})$ and the divergence theorem would then yield

$$\oint\!\!\!\oint_S \vec{\boldsymbol{A}}(\vec{\boldsymbol{r}}) \cdot \hat{\boldsymbol{n}}(\vec{\boldsymbol{r}})\,\mathrm{d}a(\vec{\boldsymbol{r}}) = \iiint_V \overrightarrow{\boldsymbol{\nabla}}_{\vec{\boldsymbol{r}}} \cdot \vec{\boldsymbol{A}}(\vec{\boldsymbol{r}}) \mathrm{d}^3\vec{\boldsymbol{r}}$$

writing $\mathrm{d}a(\vec{\boldsymbol{r}})$ as $\mathrm{d}^2\vec{\boldsymbol{r}}$ for the surface integral we can rewrite the same divergence theorem as

$$\oint\!\!\!\oint_S \vec{\boldsymbol{A}}(\vec{\boldsymbol{r}}) \cdot \hat{\boldsymbol{n}}(\vec{\boldsymbol{r}})\mathrm{d}^2\vec{\boldsymbol{r}} = \iiint_V \overrightarrow{\boldsymbol{\nabla}}_{\vec{\boldsymbol{r}}} \cdot \vec{\boldsymbol{A}}(\vec{\boldsymbol{r}}) \mathrm{d}^3\vec{\boldsymbol{r}}$$

Let us recal that for an arbitrary charged body inside a volume V enclosed by a surface S with a charge density function ρ the integral form of Gauss's law states that

$$\oint\!\!\!\oint_S \overrightarrow{\boldsymbol{E}}_{\boldsymbol{\rho}}(\vec{\boldsymbol{r}}) \cdot \mathrm{d}\vec{\boldsymbol{a}}(\vec{\boldsymbol{r}}) = \iiint_V \frac{\rho(\vec{\boldsymbol{r}}')}{\varepsilon_0} \mathrm{d}^3\vec{\boldsymbol{r}}$$

We can now use Gauss's divergence theorem on the electric field $\overrightarrow{\boldsymbol{E}}_{\boldsymbol{\rho}}(\vec{\boldsymbol{r}})$ and write

$$\Rightarrow \iiint_V \overrightarrow{\boldsymbol{\nabla}}_{\vec{\boldsymbol{r}}} \cdot \overrightarrow{\boldsymbol{E}}_{\boldsymbol{\rho}}(\vec{\boldsymbol{r}}) \mathrm{d}^3\vec{\boldsymbol{r}} = \iiint_V \frac{\rho(\vec{\boldsymbol{r}})}{\varepsilon_0} \mathrm{d}^3\vec{\boldsymbol{r}}$$

$$\Rightarrow \iiint_V \left\{ \overrightarrow{\boldsymbol{\nabla}}_{\vec{\boldsymbol{r}}} \cdot \overrightarrow{\boldsymbol{E}}_{\boldsymbol{\rho}}(\vec{\boldsymbol{r}}) - \frac{\rho(\vec{\boldsymbol{r}})}{\varepsilon_0} \right\} \mathrm{d}^3\vec{\boldsymbol{r}} = 0$$

Now, as the surface S was chosen arbitrarily and so the volume enclosed by this surface V can also be considered arbitrarily chosen. The statement above must hold for all volumes which can contain the charged body. Therefore, the above statement can only hold for all such volumes if

$$\left\{ \overrightarrow{\boldsymbol{\nabla}}_{\vec{\boldsymbol{r}}} \cdot \overrightarrow{\boldsymbol{E}}_{\boldsymbol{\rho}}(\vec{\boldsymbol{r}}) - \frac{\rho(\vec{\boldsymbol{r}})}{\varepsilon_0} \right\} = 0$$

for any position $\vec{\boldsymbol{r}}$ inside that volume containing the charged body. This yields the following equation, which is the differential form of Gauss's law

$$\overrightarrow{\boldsymbol{\nabla}}_{\vec{\boldsymbol{r}}} \cdot \overrightarrow{\boldsymbol{E}}_{\boldsymbol{\rho}}(\vec{\boldsymbol{r}}) = \frac{\rho(\vec{\boldsymbol{r}})}{\varepsilon_0}$$

4. The curl of the electric field

We have seen the differential form of Gauss's law provides us with an equation for the divergence of the electric field. It would be helpful if we could also have an equation for the curl of the electric field at a position $\vec{\boldsymbol{r}}$. Let us consider the electric field at position $\vec{\boldsymbol{r}}_{\boldsymbol{B}}$ generated by a point charge Q_A located at position $\vec{\boldsymbol{r}}_{\boldsymbol{A}}$

$$\overrightarrow{\boldsymbol{E}}_{\boldsymbol{Q_A}}(\vec{\boldsymbol{r}}_{\boldsymbol{B}}) = \frac{1}{4\pi\varepsilon_0} Q_A \frac{\vec{\boldsymbol{r}}_{\boldsymbol{B}} - \vec{\boldsymbol{r}}_{\boldsymbol{A}}}{|\vec{\boldsymbol{r}}_{\boldsymbol{B}} - \vec{\boldsymbol{r}}_{\boldsymbol{A}}|^3}$$

$$
\begin{aligned}
&= \frac{1}{4\pi\varepsilon_0} Q_A \left\{ -\vec{\nabla}_{\vec{r}_B} \left(\frac{1}{|\vec{r}_B - \vec{r}_A|} \right) \right\} \\
&= -\vec{\nabla}_{\vec{r}_B} \left\{ \frac{1}{4\pi\varepsilon_0} Q_A \left(\frac{1}{|\vec{r}_B - \vec{r}_A|} \right) \right\}
\end{aligned}
$$

We can therefore write the electric field as the gradient of a scalar function. Now, taking the electric fields curl we get

$$
\begin{aligned}
\vec{\nabla}_{\vec{r}_B} \times \vec{E}_{Q_A}(\vec{r}_B) &= -\vec{\nabla}_{\vec{r}_B} \times \vec{\nabla}_{\vec{r}_B} \left\{ \frac{1}{4\pi\varepsilon_0} Q_A \left(\frac{1}{|\vec{r}_B - \vec{r}_A|} \right) \right\} \\
&= 0
\end{aligned}
$$

We can repeat the same calculation for the electric field generated by an arbitrary charged body with a charge density ρ. The electric field at a position $\vec{r}$ can be written as

$$
\begin{aligned}
\vec{E}_\rho(\vec{r}) &= \iiint_V \frac{1}{4\pi\varepsilon_0} \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3} \rho(\vec{r}') \mathrm{d}^3\vec{r}' \\
&= \iiint_V \frac{1}{4\pi\varepsilon_0} \left\{ -\vec{\nabla}_{\vec{r}} \left(\frac{1}{|\vec{r} - \vec{r}'|} \right) \right\} \rho(\vec{r}') \mathrm{d}^3\vec{r}' \\
&= -\vec{\nabla}_{\vec{r}} \left\{ \iiint_V \frac{1}{4\pi\varepsilon_0} \left(\frac{1}{|\vec{r} - \vec{r}'|} \right) \rho(\vec{r}') \mathrm{d}^3\vec{r}' \right\}
\end{aligned}
$$

We see in this case too, the electric field can be written as the gradient of a scalar quantity. The curl of the electric field now can be written as

$$
\begin{aligned}
\vec{\nabla}_{\vec{r}} \times \vec{E}_\rho(\vec{r}) &= -\vec{\nabla}_{\vec{r}} \times \vec{\nabla}_{\vec{r}} \left\{ \iiint_V \frac{1}{4\pi\varepsilon_0} \left(\frac{1}{|\vec{r} - \vec{r}'|} \right) \rho(\vec{r}') \mathrm{d}^3\vec{r}' \right\} \\
&= 0
\end{aligned}
$$

So, for stationary charges the electric field at position $\vec{r}$(not including the position of the charge) obeys the following

$$
\vec{\nabla} \times \vec{E}(\vec{r}) = 0
$$